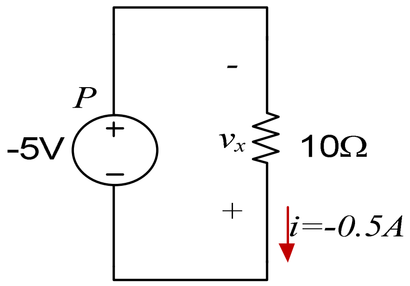
EBN 111 Klastoets / Class Test # 1 (21 Feb 2014)
15 Marks, 30 minutes

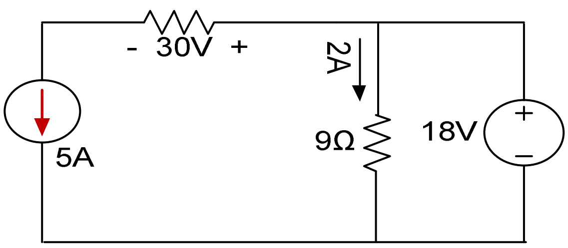
1. Hoeveel energie word in 'n 12V kar battery van 60Ah gestoor? / *How much energy is stored in a 12 V car battery rated 60Ah?* (1)

2. Die lading deur 'n element val van 10mC tot 2mC tussen tyd 0ms en 5ms. Wat is die stroom deur element by 3ms? / *The charge through an element drops from 10mC to 2mC between time 0ms and 5ms. What is the current through the element at 3ms?* *(1)*

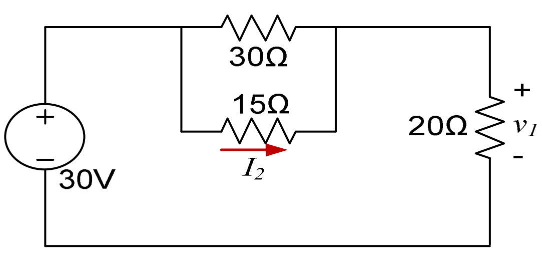
3. Bepaal die spanning v_x oor die resistor, asook die drywing P van die spanningsbron. / *Determine the voltage v_x across the resistor and the power (P) associated with voltage source.* (2)



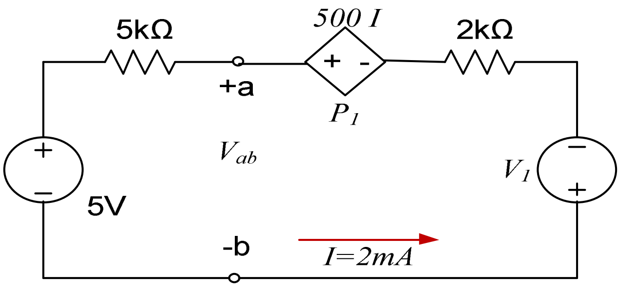
4. Bepaal die drywing geassiosieer met die 5A stroombron. / *Calculate the power associated with 5A current source.* (2)



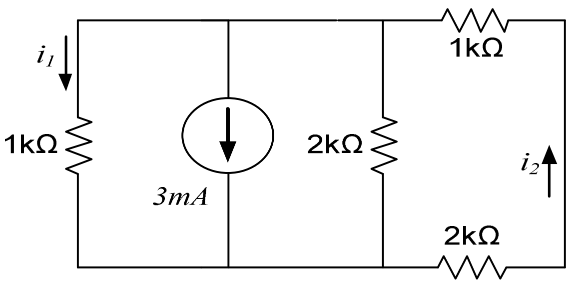
5. Bepaal v_1 en I_2. / *Calculate v_1 and I_2.* (2)



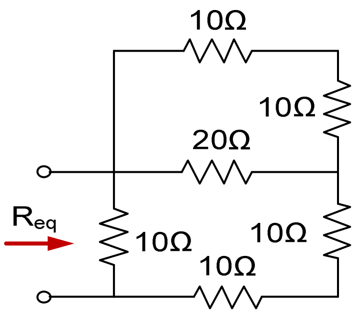
6. Bepaal V_{ab} en V_1. / *Determine V_{ab} and V_1?* (2)



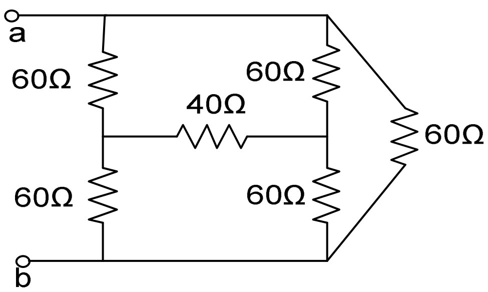
7. Bepaal i_1 and i_2. / *Find i_1 and i_2.* (2)



8. Bepaal die ekwivalente weerstand R_{eq}. / *Calculate the equivalent resistance R_{eq}.* (1)



9. Bepaal die ekwivalente weerstand R_{ab} tussen terminale *a* en *b*. / *Find equivalent resistance R_{ab} across nodes a and b.* (2)



<u>EBN111 Klastoets / Class Test # 1 (21 Feb 2014)</u>
<u>ANTWOORDBLAD / ANSWER SHEET</u>

Student besonderhede / *Student details*			
Van: *Surname*:	MEMO	Voorletters *Initials:*	
Graadkursus: *Degree Course:*		Studentenommer: *Student number:*	

MARKS	
TOTAAL/TOTAL	

VRAAG – QUESTION 1		
1	w= 2.592MJ	
2	I= -1.6A	
3	v_x= 5V	P= -2.5W
4	P_{5A}= -60W	
5	v_1= 20V	I_2= 0.67A
6	V_{ab}= 15V	V_1= -18V
7	i_1= -1.64mA	i_2= 0.54mA
8	R_{eq}= 7.5ohm	
9	R_{ab}= 30ohm	